

EC 9560: DATA MINING
INDIVIDUAL PROJECT
2026

NAME : NAFRAS M.N.A.
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Comprehensive Study on the Selected Dataset

1. Project and Dataset Recap

- Project Heading: Laptop Price Prediction using Hardware Specifications
- Dataset Used: Laptop Price Dataset (laptop_price.csv)
- Total Samples: 1,303 laptops
- Total Features: 13 columns (12 predictive features + 1 target variable, Price_euros)
- No missing values were found in any of the 13 columns.

2. Overview of Categorical Features

Before analysing numeric relationships, the categorical composition of the dataset was examined to understand how the samples are distributed across brands, laptop types, and operating systems.

- Company: 19 unique manufacturers. Dell (297) and Lenovo (297) are the most represented brands, followed by HP (274) and Asus (158). Premium/niche brands such as Apple (21), Microsoft (6), and Huawei (2) are comparatively rare.
- TypeName: Notebook is the dominant category (727 laptops, 55.8%), followed by Gaming (205), Ultrabook (196), 2 in 1 Convertible (121), Workstation (29), and Netbook (25).
- OpSys: Windows 10 dominates the dataset (1,072 laptops, 82.3%), followed by No OS (66), Linux (62), Windows 7 (45), Chrome OS (27), and macOS/Mac OS X (21 combined).

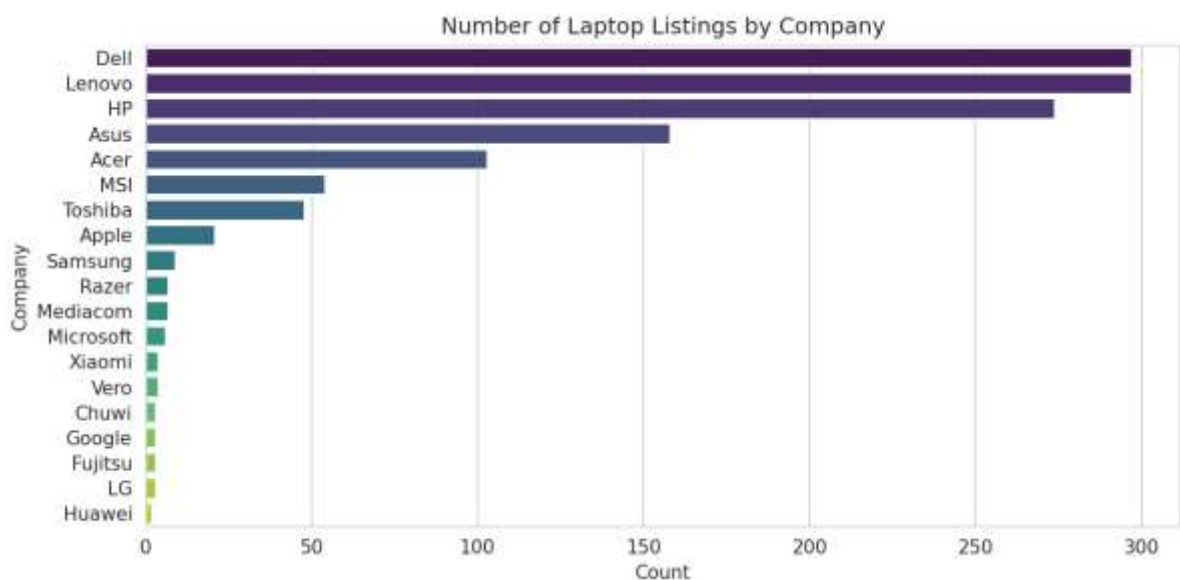


Figure 1: Number of laptop listings by manufacturer.

3. Distribution Analysis of Numeric Features

Ram and Weight were converted to purely numeric form (Ram_GB, Weight_kg) for this analysis; this conversion is finalised formally during the pre-processing stage in Lab 3.

Summary Statistics:

- Price_euros: mean = 1123.69, median = 977.00, std = 699.01, min = 174.00, max = 6099.00
- Ram_GB: mean = 8.38, median = 8, min = 2, max = 64

- Weight_kg: mean = 2.04 kg, median = 2.04 kg, min = 0.69 kg, max = 4.70 kg
- Inches: mean = 15.02, median = 15.6, min = 10.1, max = 18.4



Figure 2: Distribution (left) and boxplot (right) of Price_euros.

The target variable Price_euros is right-skewed (skewness ≈ 1.52), with the bulk of laptops priced between €400 and €1,500 and a long tail of high-end machines extending beyond €4,000. The boxplot confirms a considerable number of upper-end outliers, which is expected given the presence of premium gaming and workstation laptops in the dataset. This skew suggests that a log-transformation of Price_euros may be beneficial prior to model training.

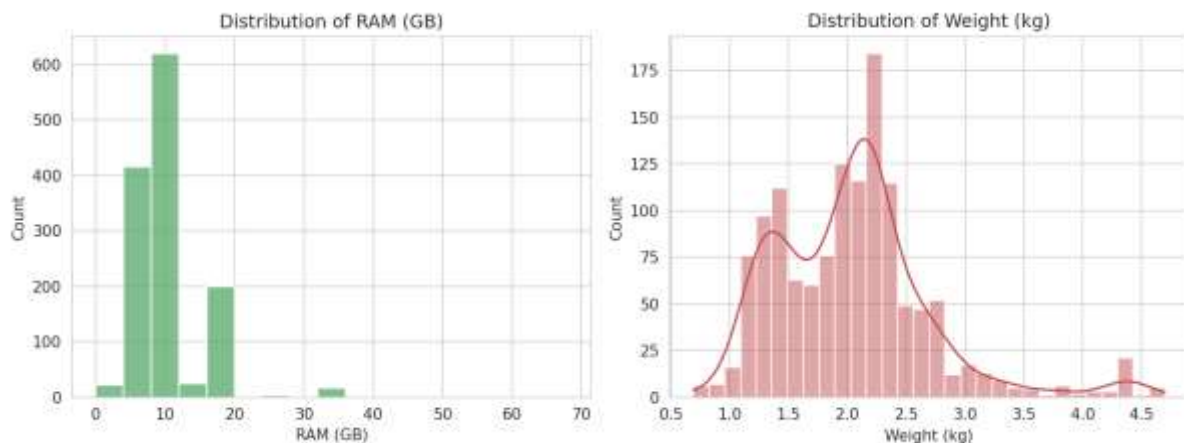


Figure 3: Distribution of RAM (left) and Weight (right).

RAM is heavily concentrated at the 4GB and 8GB tiers, which together account for the majority of listings, producing a strong right skew (skewness ≈ 2.69) due to a small number of 32GB/64GB high-performance machines. Weight is fairly close to a bell shape with mild right skew (skewness ≈ 1.14), most laptops weighing between 1.5 kg and 2.3 kg, with lightweight ultrabooks and heavier gaming/workstation laptops forming the tails.

4. Price Variation Across Categories

Workstation and Gaming laptops command the highest median prices, consistent with their higher-end hardware specifications, while Netbooks are the cheapest category by a clear margin. Notebook, the

largest category by volume, shows the widest spread of prices, indicating that it spans both budget and premium hardware configurations.

Laptops with Intel CPUs and Nvidia/AMD discrete GPUs tend to command higher median prices than those with Samsung or ARM-based processors, or with integrated graphics, reflecting the performance premium typically associated with dedicated processing hardware.

5. Correlation Analysis

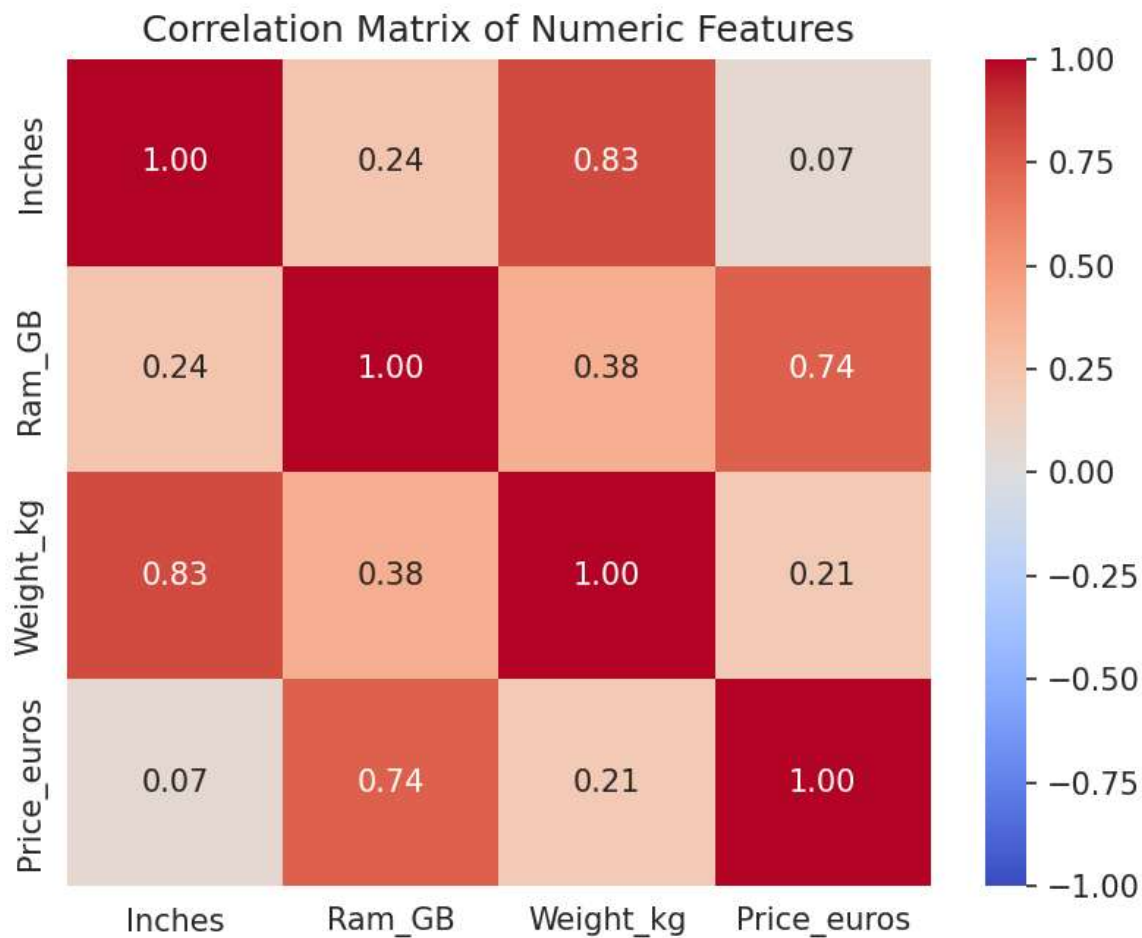


Figure 6: Correlation matrix of numeric features.

Correlation Matrix (Pearson's r):

- Price_euros vs Ram_GB: $r = 0.743$ (strong positive correlation) — RAM capacity is the strongest linear numeric predictor of price among the raw numeric fields.
- Price_euros vs Weight_kg: $r = 0.210$ (weak positive correlation).
- Price_euros vs Inches: $r = 0.068$ (negligible correlation) — screen size alone has little linear relationship with price.
- Inches vs Weight_kg: $r = 0.828$ (strong positive correlation) — larger-screen laptops are, unsurprisingly, considerably heavier.



Figure 7: Scatter plots of RAM, Weight, and Screen Size against Price.

The scatter plots reinforce the correlation findings: RAM shows a clear upward trend with price, weight shows a weak and noisy relationship, and screen size shows almost no discernible linear pattern with price. This indicates that categorical and textual fields such as Cpu, Gpu, and Memory — which have not yet been decomposed into numeric features — are likely to carry substantial additional predictive information that raw numeric correlation cannot capture.

6. Key Observations Going into Pre-processing

- The target variable Price_euros is right-skewed; a log transformation will be considered before model training.
- Ram_GB is the strongest linear predictor among the numeric fields and should be preserved as-is after conversion from string to numeric.
- Inches and Weight_kg are highly collinear ($r = 0.828$); this will be considered during feature selection in Lab 4 to avoid redundancy.
- Cpu, Gpu, and Memory contain rich compound information (e.g. clock speed, storage type and capacity) that must be decomposed into separate numeric/categorical sub-features in Lab 3.
- Categorical variables (Company, TypeName, OpSys, Cpu brand, Gpu brand) show visible separation in price and will require encoding (One-Hot or Label Encoding) before modelling.